

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **RTL modules for high-speed digital architectures** using **SystemVerilog** and **state machine** design patterns, applying timing closure and synthesis best practices to meet clock domain constraints in multi-rate enterprise systems; conducted **digital design verification** via simulation tools and achieved **100% timing margin** across all synthesized IP blocks.
- Collaborated daily with **cross-functional design, verification, and back-end engineering teams** in fast-paced production environments; led **design reviews** and technical requirement analysis with upstream stakeholders (architects, BAs, operations teams) to translate functional specifications into configurable RTL implementations, reducing design-to-verification cycle time by **18%**.
- Developed **configurable IP delivery** frameworks for multi-port, multi-rate digital cores using **Verilog** and **SystemVerilog**; architected parameterized RTL to enable flexible instantiation across different speed grades and port counts, supporting **10+ distinct system configurations** and eliminating manual code duplication.
- Executed **synthesis and timing closure workflows** using industry-standard EDA tools; debugged critical path violations and applied optimization techniques (pipelining, retiming, logic restructuring) to meet aggressive frequency targets, achieving **95% first-pass timing closure rate** across **8 silicon tapeouts**.
- Applied **digital design fundamentals** including **clocking architecture, reset distribution, metastability handling**, and **CDC (Clock Domain Crossing)** best practices in high-speed interconnect IP; validated designs using **assertion-based verification** and formal property checking, identifying and preventing **12 critical timing and functional edge cases** pre-silicon.
- Supported **FPGA implementation** and rapid prototyping of RTL designs using industry simulation and synthesis tools; participated in **IEEE 802.3 standards committee** discussions and contributed to specification alignment activities, ensuring RTL implementation compliance with emerging high-speed Ethernet protocols and architectural requirements.

SKILLS

Digital Design & RTL – Verilog, SystemVerilog, RTL coding, state machines, timing closure, synthesis, FPGA implementation.

Design Tools & EDA – Xcelium, VCS, ModelSim, Vivado, synthesis tools, formal verification, assertion-based verification.

High-Speed Ethernet & Networking – Ethernet standards (IEEE 802.3), MAC (Media Access Control), PCS (Physical Coding Sublayer), FEC (Forward Error Correction), high-speed signal integrity, multi-rate architectures, 10Gb through 1.6Tb IP design.

Hardware Design Methodology – Digital design best practices, clocking architecture, CDC (Clock Domain Crossing), metastability handling, reset distribution, configurable IP delivery, timing analysis, logic optimization.

Engineering Practices – Cross-functional collaboration, design reviews, requirements analysis, verification workflows, version control (Git), technical documentation, Agile/Scrum ceremonies.

Additional Technical Skills – Linux, shell scripting, Python scripting, SQL (PostgreSQL, MySQL, Oracle), AWS (EC2, S3, Lambda), Docker, CI/CD (Jenkins), Jira, Confluence.

Domain Expertise – Electronic design, computational software, silicon delivery, standards committee participation (IEEE 802.3, Ultra Ethernet Consortium), AI infrastructure Ethernet IP development.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India

01/2021

PROJECTS

Multi-Rate Ethernet MAC & PCS RTL Architecture (HCL Technologies)

- Architected and coded **parameterized RTL for multi-rate Ethernet MAC (Media Access Control) and PCS (Physical Coding Sublayer) controllers** supporting 10Gb, 25Gb, 100Gb, and 400Gb speed grades using **SystemVerilog** with configurable port counts and frame-size flexibility; reduced design code duplication by **40%** through reusable, synthesizable module templates.
- Implemented **FEC (Forward Error Correction) logic** with **state machine**-based syndrome calculation and error correction pathways; achieved **99.8% bit-error-rate mitigation** under high-noise simulation conditions and verified functional correctness across **50+ corner-case test scenarios** using **Xcelium** and **SystemVerilog assertions**.
- Applied rigorous **timing closure methodology** and **synthesis optimization techniques** to meet **1.6Tb Ethernet frequency targets**; performed iterative **static timing analysis** across all process corners, voltage, and temperature (PVT) variations, achieving **98% first-pass timing closure** and zero timing violations in post-synthesis netlists.
- Designed **clock domain crossing (CDC) logic** and **metastability handling circuits** for multi-port architectures operating asynchronously; validated CDC correctness using **formal property verification** and **specialized CDC simulators**, eliminating potential synchronization failures in **7 critical cross-domain data paths**.
- Collaborated with **verification engineers** to define comprehensive RTL testbenches and stimulus generation; supported **FPGA rapid prototyping** on Xilinx platforms for pre-silicon validation and design demonstrations, reducing field issues by **25%** and accelerating time-to-market by **3 months**.

High-Speed Ethernet Configurable IP Delivery Framework (HCL Technologies)

- Developed **configurable RTL delivery framework** for multi-port Ethernet IP cores, enabling parametric instantiation across **10+ distinct customer configurations** (speed grades, port counts, FEC modes) without code duplication; streamlined customer onboarding by **6 weeks** through self-documenting, template-driven RTL generation.
- Executed **synthesis and timing closure workflows** across **8 silicon tapeouts**, achieving **95% first-pass timing success rate**; applied **logic optimization**, pipelining, and retiming strategies to resolve critical-path violations and maintain **100% yield** across all taped-out configurations.
- Participated in **IEEE 802.3 standards committee** design reviews, translating emerging Ethernet specifications into compliant RTL; ensured architectural alignment with **Ultra Ethernet Consortium** requirements, reducing specification-to-implementation gaps by **30%** and establishing Cadence Cork as a recognized standards contributor.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)